



AnnoBot – Experiment Results

Recap: previous meeting

- ▶ **Experiments:** Meeting Room (TV) and, newly added, Kitchen (Fridge, Microwave).
- ▶ **Fact $\rightarrow$ Case:** a collection of sensor observations, contextual information, and semantic annotations relevant to the inquiry goal.
- ▶ **Signature extraction $\rightarrow$ Cycle detection:** an algorithmic rule that scans sensor traces and returns the temporal boundaries $[t_{\text{start}}, t_{\text{end}}]$ of an active service episode or appliance cycle.

Experiment setup: $E = \langle I, T \rangle$

	Kitchen	Meeting Room
Period	28/09 – 28/10/2026 (1 month)	same
Ask window	24/7 (testing only; realistically 9h–21h)	same
Budget	10 asks/day	10 asks/day
IL timestep	60 minutes	30 minutes

Timezone: Asia/Ho_Chi_Minh.



Demo: Experiment list



Kitchen: Fridge cycles & microwave use

Asks every 60 min

10 asks/day · 00:00–23:59



Meeting room: Usage & comfort

Asks every 30 min

10 asks/day · 00:00–23:59

Inquiries: $\iota = \langle Q, S, I, D, A, \Gamma \rangle$

Inquiry	Question Q	Sensors S	Detection D	Annotation A
Fridge	How regular is the compressor cycle, and how much energy does each cycle use?	fridge plug	power_cycle, learned thresholds	none (<i>annotation-free</i>)
Microwave	When is it used, and what for?	microwave plug	power_cycle, learned thresholds	what required
Session	When is the meeting room used, and for how long?	TV plug, temperature, humidity	power_cycle, UK-DALE profile	who, what required
Comfort	Do temperature and humidity stay in the comfort band during meetings?	temperature, humidity	climate_ observation	why, how, performance required

Inquiry goals and indicators

Inquiry	Goal Γ	Indicators I
Fridge	Confirm the detector draws consistent on/off boundaries across many repetitions (not anomaly detection).	energy, duration, mean power, peak power
Microwave	Handle a sparse, short, noisy signal, and sort each burst into one of four uses.	energy, duration, peak power
Session	Measure real meeting-room occupancy rather than inferring it from the TV alone.	energy, duration, mean power
Comfort	Trade comfort against energy, and surface AC left running when nobody is in the room.	avg. temperature, avg. humidity



Demo: Meeting-room inquiries

Inquiries

+ Add inquiry

Define target questions, monitored sensors, cycle detection rules, and occupant response expectations.

When is the meeting room used, and for how long?

Fact  

Goal: Measure real meeting-room occupancy rather than inferring it from the TV alone.

Assigned Sensors

 Z4 Meeting Plug TV Power  Z4 Meeting Climate Air temperature

 Z4 Meeting Climate Humidity

Detection Rule

Power Cycle Detection P_on 10 W · P_off 5.16 W

Occupant Response

who what required upon case trigger

Do temperature and humidity stay in the comfort band during meetings, and is the AC left running outside them?

Goal: Trade comfort against energy, and surface AC left running when nobody is in the room.

Performance  

Assigned Sensors

 Z4 Meeting Climate Air temperature  Z4 Meeting Climate Humidity

Detection Rule

climate_observation (Thresholds auto-calibrated from runtime telemetry)

Occupant Response

how why performance required upon case trigger

Cycle detection: adaptive ON/OFF thresholds

- ▶ **Goal:** find $[t_{\text{start}}, t_{\text{end}}]$ of each appliance cycle from raw power, with no hand-tuned thresholds.
- ▶ **Learn thresholds from data:** pick the power level that best separates the OFF (standby) and ON (active) readings [1], on $\ln(1 + P)$ since power is multiplicative and right-skewed [4].
- ▶ **Hysteresis:** two thresholds $P_{\text{on}} > P_{\text{off}}$ above a noise floor, so a cycle does not flicker around one value.
- ▶ **Clean up:** drop too-short cycles, merge short pauses [2]. Thresholds directly determine the ON/OFF labels [3].

Why not Median + $k \cdot \text{MAD}$? It assumes OFF dominates, but a fridge is ON $\sim 70\%$ of the time, so the median lands inside ON and cycles are missed.

Learned thresholds

Appliance	$P_{\text{on}} / P_{\text{off}}$	Max gap	Duration	Source
Fridge	$\approx 9.2 / 5.2 \text{ W}$	0 min	3–120 min	learned
Microwave	$\approx 24 / 10.8 \text{ W}$	2 min	1–15 min	learned
TV	$10 / \approx 5.2 \text{ W}$	5 min	$\geq 5 \text{ min}$	UK-DALE profile [5]

Learned thresholds are re-estimated on each run, so they vary slightly (fridge P_{on} 9.1–9.4 W; microwave 24.0 W).

Demo: Thresholds recorded with a case

Case Details & Evidence

Case 30/09 11:56 - 30/09 12:14. View recorded annotations and detection parameters.

Thresholds used

otsu-log

❖ Detection rule updated at this cycle

heartbeat_h: 5.17 → 0.08 **gap_method:** too-few-gaps → bimodal(ratio=5.5) **gap_hold_h:** 5.68 → 0.09
p_off_w: 5.21 → 5.16

Turn-on (P_on)	Turn-off (P_off)	Max gap	Min duration
10 W	5.2 W	5 min	5 min
		Algorithm: bimodal(ratio=5.5)	

Measurements & Energy cross-check

Measured power and two independent energy sources: the engine integrating power, and the cumulative meter. Agreement means the boundary is right.

Peak (instant)	Integrated power	Cumulative meter	Drift
127 W	34.9 Wh	—	—
		Meter step 10 Wh, case too small for cross-check	No meter reference data

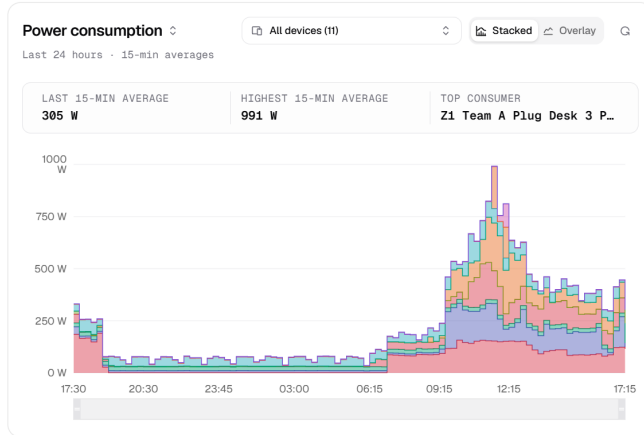
Overview: 3-day live run

Inquiry	Cases	Status	Avg. duration	Energy
Fridge	48	48 annotation-free	54 min (median)	~39 Wh / cycle
Microwave	5	5 complete	7.0 min	~99 Wh / use
TV session	3	3 complete	19 min	~41 Wh / session
Comfort	0	no detector yet	—	—

Mean, peak, and energy come from the integrated raw events ($E_{\text{cycle}} = \frac{1}{3600} \int_{t_{\text{start}}}^{t_{\text{end}}} P(t) dt$).



Demo: Power dashboard



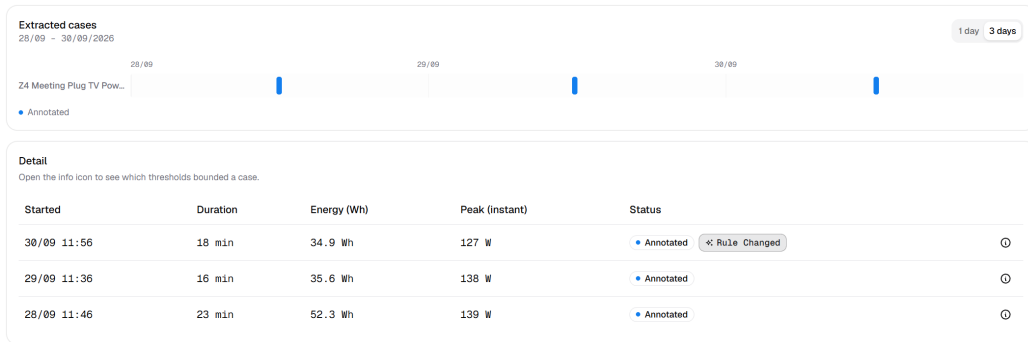
Meeting Room: a recurring lunch-break routine

Occupants turn on the TV to watch YouTube while eating lunch, on three consecutive days between **11:35 and 12:15**.

Date	Time	Dur.	Mean	Peak	Energy	Who	What
28/09	11:46–12:09	23 min	136 W	139 W	52.3 Wh	many people	eating and watching YouTube on TV
29/09	11:36–11:52	16 min	133 W	138 W	35.6 Wh	6 people	eating and watching TV
30/09	11:56–12:14	18 min	116 W	127 W	34.9 Wh	many people	eating and watching TV on YouTube



Demo: Extracted cases



Demo: Real-life event: lunch with the TV on





Demo: Annotation chat

← When is the meeting room used, and for how long? »

✓ Annotated

✓ who: 6 people

✓ what: people eating a...

(2/2)

ⓘ Details ▾

• thought for 2s · telemetry verified

We observed the Z4 Meeting Plug TV Power activity in the meeting room during Sep 29, 11:36-11:52. Who was involved in this activity?

(x)

> 6 people

04:04 PM

• thought for 2s · telemetry verified

We still need to determine what was happening during this time. What specifically took place?

(x)

> people eating and watching TV

04:05 PM

• thought for 2s · telemetry verified

Thanks, I've recorded that for this session.

(x)

> Type your reply or pick a suggestion above...

↑

Enter to send, Shift+Enter for newline

• AnnoBot

Kitchen: microwave

All 5 uses were annotated “*reheating a meal*”, with peaks around 1.2 kW.

Date	Time	Duration	Peak	Energy
28/09	11:40–11:47	7 min	1,198 W	94.9 Wh
28/09	12:05–12:13	8 min	1,208 W	103.8 Wh
29/09	11:24–11:39	15 min	1,228 W	211.1 Wh
30/09	11:38–11:41	3 min	1,240 W	50.3 Wh
30/09	12:00–12:02	2 min	1,219 W	33.1 Wh

Kitchen: fridge (annotation-free)

48 cycles in ~ 2.7 days (≈ 18 per day), all handled without asking the user.

Metric	45 regular cycles
Duration	34–93 min (median 54)
Mean power	39–103 W (median 43)
Peak power	52–139 W
Cycle energy	28–79 Wh (median 39)

Outliers

3 cycles (119, 158, and 305 min) exceed the duration limit and carry the duration-over-max flag.

Limitations

1. **No timestep resampling & discretization:** cases are not yet projected onto a 15/30/60-min grid or binned into Low/Medium/High, so X_T for the IL trigger is not built.
2. **Appliance-only detection:** the engine supports only power_cycle; the comfort inquiry produced 0 cases.
3. **No cooperative quality control:** users cannot confirm, edit, or reject bot-inferred labels.
4. **No occupancy sensors:** without PIR or door contacts, who depends on user input.
5. **Abnormally long fridge cycles** (3 of 48) are not yet split or explained.

Future work

Timestep discretization

Resample cases onto a standard grid (15/30/60 min) to build X_T , then bin values into Low/Medium/High for the IL trigger.

Climate binding

Attach temperature and humidity to each TV interval $[t_{\text{start}}, t_{\text{end}}]$ and add comfort-band rules.

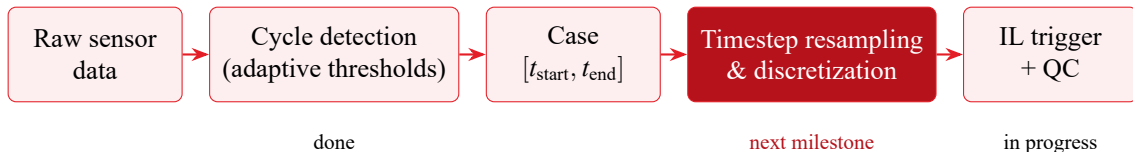
Cooperative QC

Web UI to Confirm / Edit / Reject / Defer bot-filled labels.

Occupancy sensing

PIR, door contact, and CO₂ to estimate headcount (who).

Pipeline and roadmap



- ▶ **Next step, after a case exists:** *temporal resampling* projects cases of different lengths onto a standard time grid (15/30/60 min) to build the analysis matrix X_T ; *semantic binning* turns numeric values into Low / Medium / High.
- ▶ **Why:** the IL trigger (neighbourhood density) needs time-aligned feature vectors to compare cases and decide whether to ask the user.

References

- [1] N. Otsu, “A threshold selection method from gray-level histograms,” *IEEE Trans. Syst., Man, Cybern.*, vol. 9, no. 1, pp. 62–66, 1979.
- [2] J. Kelly and W. Knottenbelt, “Neural NILM: Deep neural networks applied to energy disaggregation,” in *Proc. ACM BuildSys*, 2015, pp. 55–64.
- [3] D. Precioso and D. Gómez-Ullate, “Thresholding methods in non-intrusive load monitoring,” *J. Supercomputing*, vol. 79, pp. 14039–14062, 2023.
- [4] W. A. H. Jumiawi and A. El-Zaart, “Improvement in the between-class variance based on lognormal distribution for accurate image segmentation,” *Entropy*, 2022.
- [5] J. Kelly and W. Knottenbelt, “The UK-DALE dataset, domestic appliance-level electricity demand and whole-house demand from five UK homes,” *Scientific Data*, vol. 2, 150007, 2015.